

MindRef: Mimicking Human Memory for Hierarchical Reference Retrieval with Fine-Grained Location Awareness

Anonymous ACL submission

Abstract

When completing knowledge-intensive tasks, humans sometimes need an answer and a corresponding reference passage for auxiliary reading. Previous methods required obtaining pre-segmented article chunks through additional retrieval models. This paper explores leveraging the parameterized knowledge stored during the pre-training phase of large language models (LLMs) to recall reference passage from any starting position independently. We propose a two-stage framework that simulates the scenario of humans recalling easily forgotten references. Initially, the LLM is prompted to recall document title identifiers to obtain a coarse-grained document set. Then, based on the acquired coarse-grained document set, it recalls fine-grained passage. In the two-stage recall process, we use constrained decoding to ensure that content outside of the stored documents is not generated. To increase speed, we only recall a short prefix in the second stage, and then locate its position to retrieve a complete passage. Experiments on KILT knowledge-sensitive tasks have verified that LLMs can independently recall reference passage locations in various task forms, and the obtained reference significantly assists downstream tasks.

1 Introduction

Knowledge-intensive tasks rely heavily on large knowledge sources (Petroni et al., 2021). Traditional methods often use retrieval models to find relevant passages from resources like Wikipedia for tasks such as question answering (Izacard and Grave, 2021). However, limitations exist with both sparse (lack of semantic depth) and dense retrieval (limited interaction between question and passage representations) (Khattab et al., 2021). Generative retrieval methods, leveraging models’ generative abilities for deeper interaction with knowledge sources, are gaining popularity (Tay et al., 2022). However, Current retrieval methods require pre-segmented passages, limiting reference flexibility

like human memory. We ask: "Can LLMs bypass chunking to recall references from any position?"

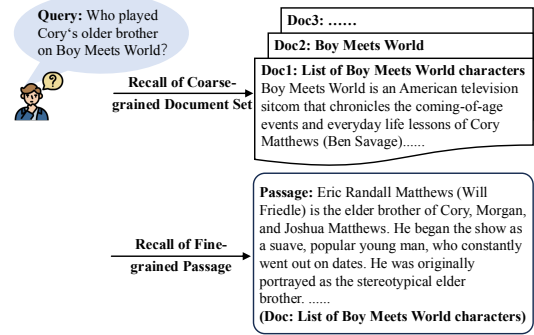


Figure 1: Human recall of forgotten information often involves a two-step process: recalling memorable documents first, then locating the specific passage within.

Leveraging the capabilities of LLMs, we propose **MindRef**, a two-stage framework for flexible passage retrieval. Inspired by human recall, we first prompt the LLM to recall relevant document titles, guided by a Trie (Cormen et al., 2022). Then, using an FM-index (Ferragina and Manzini, 2000) built from the retrieved documents, the LLM recalls specific passages with flexible starting points. A weighted score combines both stages for final reference selection. To enhance efficiency, MindRef retrieves passages by recalling only a short prefix. The LLM generates this prefix, which is located within the documents using the FM-index and KMP algorithm. Ultimately, the algorithm identifies a longer passage as the final reference. This approach allows LLMs to access and retrieve natural references from articles of any length without relying on additional retrieval models or pre-segmentation, offering both flexibility and efficiency.

Extensive experiments on 6 KILT benchmark tasks (Petroni et al., 2021) demonstrate the effectiveness of **MindRef**, enabling open-source LLMs like LLaMA (Touvron et al., 2023a) and LLaMA-2

(Touvron et al., 2023b) to retrieve documents and passages, improving downstream task performance effectively. Key contributions include:

1) We propose MindRef, a cognitively-aligned retrieval framework that formalizes the ‘document-to-detail’ mechanism of human memory into neural architectures for the first time. 2) Breaking away from pre-chunking paradigms, our method achieves chunkless reference localization through joint Trie-FMIndex constrained decoding, enabling retrieval from arbitrary document positions. 3) The SPRL co-optimization strategy delivers 4× faster inference while maintaining 95%+ accuracy via short-prefix neural recall. 4) Extensive evaluations on 6 KILT tasks demonstrate state-of-the-art performance, with MindRef-boosted LLaMA-2 achieving 78.79% accuracy on FEVER.

2 MindRef Framework

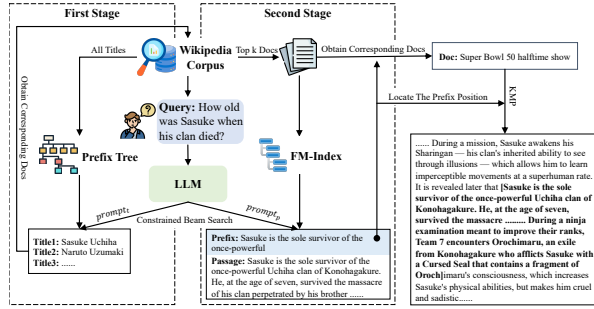


Figure 2: **MindRef** Framework. First, all Wikipedia titles are stored in a prefix tree, then the LLM is prompted to recall title identifiers under this prefix tree constraint. Subsequently, an FM-index is constructed from the top k documents obtained, and the LLM recalls reference passage under the new constraint.

In this section, we detail our two-stage framework, **MindRef** (Figure 2). In the first stage, we prompt the LLM to recall title identifiers, which serve as candidate documents for the next stage. In the second stage, the LLM is prompted to recall reference passage from the documents obtained in the first stage. To increase speed, we only recall a short prefix, then locate and extract the reference within the document. Detailed Prompt can be found in Subsection B.2.

2.1 Stage 1: Coarse-Grained Document Recall

Recalling fine-grained reference passages directly for knowledge-intensive tasks can be challenging (Subsection B.4). Therefore, we propose a two-stage process. First, we retrieve easily-recallable

documents (e.g., Wikipedia pages) by leveraging their titles as unique identifiers. Using a Trie (Section A.1) data structure (Cormen et al., 2022), we prompt the LLM to recall relevant titles, ensuring they correspond to existing pages.

Given a query x and prompt $prompt_t(x)$ (e.g., "Question: \n \n The Wikipedia title corresponding to the above question is: \n \n Title:"), the LLM generates titles, guided by the Trie. The Trie, based on previously generated tokens, restricts subsequent token choices to valid prefixes within the set of all Wikipedia titles (T), effectively guiding the LLM along valid title paths.

This first stage focuses on efficiently retrieving a set of candidate documents before proceeding to fine-grained passage retrieval within these documents in the second stage. The score for generating title t given prompt $prompt_t(x)$ is calculated using the standard implementation from the library:

$$\begin{aligned} score_1(t|prompt_t(x)) &= \frac{\log p_\theta(y_t|prompt_t(x))}{|y_t|} \\ &= \frac{\sum_{i=1}^{l_t} \log p_\theta(y_i|y_{<i}, prompt_t(x))}{l_t} \end{aligned} \quad (1)$$

where y_t represents the set of tokens in title t , l_t and $|y_t|$ represent the number of tokens used to generate the title, θ represents the model’s parameters.

2.2 Stage 2: Fine-Grained Passage Recall

Following the initial identification of relevant Wikipedia pages, we move to fine-grained passage retrieval within those documents. We employ the FM-index ((Section A.2)) constraint (Ferragina and Manzini, 2000), a space-efficient data structure enabling fast substring search and supporting retrieval from arbitrary positions.

After obtaining the top k titles and their documents (D_k), we construct a targeted FM-index specifically for D_k , reducing the search space. These FM-indexes are pre-built for all documents to avoid on-the-fly construction.

We then prompt the LLM with $prompt_p$ (e.g., "Question: \n \n The answer to the above question can be found in the following Wikipedia paragraph:\n \n Answer:") to generate a passage p . The FM-index, based on previously generated tokens, dynamically provides permissible successor tokens, guiding the LLM to generate valid passages from any position within D_k . We measure the score of the task corresponding passage by using the autore-

gressive formula to calculate the score:

$$\begin{aligned} & score_2(p|prompt_p(x)) \\ &= \frac{\log p_\theta(y_p|prompt_p(x))}{|y_p|} \\ &= \frac{\sum_{i=1}^{l_p} \log p_\theta(y_i|y_{<i}, prompt_p(x))}{l_p} \end{aligned} \quad (2)$$

where y_p represents the set of tokens in the passage p , θ is the model parameters, $|y_p|$ and l_p represent the number of tokens generating the passage, usually set between 150 to 200. To integrate information generated from both stages, we calculate the weighted sum of the scores from the first and second stages to obtain the final score under input query x :

$$\begin{aligned} score(p|x) &= \alpha * score_1(t|prompt_t(x)) \\ &+ (1 - \alpha) * score_2(p|prompt_p(x)) \end{aligned} \quad (3)$$

where $score_1(t|prompt_t(x))$ is the score of the Wikipedia page title t corresponding to the passage p , α is a hyperparameter controlling the weight of the two stages. Finally, among all recalled passages, the one with the highest $score(p|x)$ value is selected as the best reference.

2.3 Short Prefix Recall and Localization

While LLMs excel at recalling long passages, their inference speed hinders practical application. To address this, we propose Short Prefix Recall Location (SPRL), aiming to locate passages by recalling only a short prefix.

Given a question q , SPRL first prompts the LLM to generate a short prefix p_s of length l_{p_s} using the same prompt ($prompt_p$) as in the second stage, significantly reducing generation cost. Subsequently, SPRL identifies the document d containing p_s within the pre-selected document set D_k . Using the KMP algorithm, the starting position st of p_s in d is determined, and a complete reference passage $p_{final} = d[st : st + l_p]$ is extracted. The final score is calculated using Equations 2 and 3 to select the best reference. Experiments (Section 3.2.1) demonstrate that recalling only the prefix for localization yields effective results.

3 Experiments

In this section, we conduct comprehensive experiments on coarse-grained pages, fine-grained passage-level reference evaluation, and downstream tasks to validate the effectiveness of our framework. Additionally, we perform further analyses and experiments in the Appendix through Ablation Studies, Further Analysis, and Case Studies.

Table 1: Coarse-grained page-level results (R-Precision). $\star$ denotes full data training. Cyan indicates best results, and pink indicates second-best.

Method	Open-domain QA				Fact Check.	Dial.
	NQ	TriviaQA	HotpotQA	ELIS	FEVER	WoW
Contriever	34.72	34.28	26.14	11.02	55.64	29.67
BM25	26.33	31.78	41.30	6.83	52.09	28.78
DPR*	54.74	45.68	25.46	16.19	56.61	26.62
MindRef (LLaMA-7b)	54.46	57.03	44.56	15.13	76.57	52.91
MindRef (LLaMA-13b)	54.42	55.53	46.30	12.94	77.55	34.51
MindRef (LLaMA-2-7b)	56.33	56.43	46.20	14.60	77.29	49.61
MindRef (LLaMA-2-13b)	57.77	54.41	48.70	15.00	83.69	57.63

Table 2: Fine-grained passage-level results (Answer/Entity in Context for top-1 reference).

Method	Open-domain QA				Fact Check.	Dial.
	NQ	TriviaQA	HotpotQA	ELIS	FEVER	WoW
	Answer in Context			Entity in Context		
Contriever	19.28	37.21	11.16	12.48	40.48	45.15
BM25	23.65	58.87	29.45	12.01	58.33	50.36
DPR*	47.94	66.60	20.29	14.40	41.22	45.38
MindRef (LLaMA-7b)	36.87	58.48	25.55	15.99	54.85	59.40
MindRef (LLaMA-13b)	37.72	60.96	26.34	14.80	55.20	50.79
MindRef (LLaMA-2-7b)	38.07	62.88	27.55	16.85	56.23	57.79
MindRef (LLaMA-2-13b)	40.82	68.20	30.04	15.06	58.42	63.43

3.1 Experimental Setup

Datasets: Experiments were conducted on 6 knowledge-sensitive tasks from the KILT benchmark (Petroni et al., 2021) ((Section B.1)).

Evaluation Metrics: R-Precision for page-level retrieval. Answer in Context (percentage of references containing at least one gold answer) for NQ, TriviaQA, and HotpotQA. Entity in Context (percentage of references containing at least one gold entity) for other datasets. Downstream task metrics followed the official KILT implementations.

Baseline Models: Compared against BM25 (Robertson and Zaragoza, 2009), Contriever (Izacard et al., 2022), and DPR (Karpukhin et al., 2020) using KILT passage segmentation.

Implementation Details: LLaMA (Touvron et al., 2023a) and LLaMA-2 (Touvron et al., 2023b) (7b and 13b) were used for reference recall. LLaMA-2-13b served as the reading model for downstream tasks. KILT passage fragments were merged into complete documents. Beam search was used for generation (beam size 15 for the first stage, 10 for the second). FM-index was constructed with top $k = 2$ documents. Short prefix length $l_{p_s} = 16$, final reference length $l_p = 150$. Two-stage weight $\alpha = 0.9$. Greedy decoding for downstream tasks. Experiments were conducted on Tesla A100 40G GPUs.

Table 3: Downstream task results.

Method	Open-domain QA				Fact Check.	Dial.
	NQ	TriviaQA	HotpotQA	ELI5	FEVER	WoW
	EM			R-L	ACC	F1
LLaMA-2-13b	19.74	68.71	15.64	19.46	73.23	13.90
Contriever	24.78	69.25	20.34	20.71	73.61	13.96
BM25	25.84	71.49	27.23	20.48	77.54	14.02
DPR*	33.49	72.68	23.13	20.75	75.27	14.17
MindRef (LLaMA-7b)	29.78	70.18	24.61	20.60	78.10	14.47
MindRef (LLaMA-13b)	29.68	71.60	25.48	20.24	78.53	14.33
MindRef (LLaMA-2-7b)	29.89	70.04	25.55	20.50	78.04	14.48
MindRef (LLaMA-2-13b)	31.69	72.94	26.13	20.61	78.79	14.77

3.2 Experimental Results

3.2.1 Page-level Results

Coarse-grained page-level results (Table 1), demonstrate that the MindRef framework, when implemented with LLaMA-2-13b, achieves the best R-precision scores. This significantly surpasses the performance of sparse retrieval BM25 and dense retrieval Contriever in a zero-shot scenario. It also shows strong competitive power against the fully trained DPR method, especially on the WoW and FEVER datasets. This result is consistent with the hypothesis that LLMs are powerful in recalling coarse-grained title identifiers, enabling the acquisition of high-quality relevant pages that assist in the subsequent fine-grained recall stage.

3.2.2 Passage-level Results

Table 2 shows that MindRef with LLaMA-2-13b achieves top scores for fine-grained passage retrieval on 4 datasets. While improvements over DPR are less pronounced than at the page-level, this highlights the difficulty of recalling longer references. MindRef’s significant progress on FEVER and WoW demonstrates its adaptability. The positive correlation between model size/architecture (LLaMA vs. LLaMA-2) and performance suggests a link between recall ability and LLM capabilities.

3.2.3 Downstream Task Results

Downstream task results are presented in Table 3. MindRef, based on LLaMA-2-13b recalled passage, achieved the best scores of 3 downstream tasks, validating the performance of LLM recall references in downstream tasks. In the TriviaQA and HotpotQA datasets, due to the length of the questions, BM25 achieved excellent performance by obtaining more vocabulary overlap, yet MindRef still achieved comparable or better performance in most cases. The unsupervised trained Contriever performed relatively poorly across all tasks, emphasizing the crucial role of supervised training

in enhancing the performance of dense retrieval models.

4 Related Work

Traditional retrieval methods rely on sparse (TF-IDF, BM25 (Robertson and Zaragoza, 2009)) or dense (ORQA (Lee et al., 2019), DPR (Karpukhin et al., 2020)) representations. However, dual-encoder dense retrieval faces limitations due to shallow interactions (Khattab and Zaharia, 2010).

Recent work explores using LLMs to generate identifiers for retrieval, targeting page titles (De Cao et al., 2021), hierarchical paths (Tay et al., 2022), n-grams (Bevilacqua et al., 2022), multi-hop paths (Lee et al., 2022), multiple identifiers (Li et al., 2023b), or a two-stage approach with passages and URLs (Ren et al., 2023). These methods, however, often retrieve predefined text segments, hindering flexible retrieval from arbitrary positions.

Leveraging LLMs to directly generate relevant knowledge (Fang et al., 2022) or augmenting models with LLM-generated context (Yu et al., 2023) has shown promise. However, hallucination remains a challenge (Li et al., 2023a).

5 Conclusion

This paper introduces **MindRef**, a framework utilizing LLMs to independently recall reference passages for knowledge-sensitive tasks. Mimicking human information-seeking behavior, the LLM first recalls relevant document pages, and then locates specific passages within them. Beam search, constrained by Trie and FM-index structures, ensures that recalled content is a subset of existing texts. This framework is adaptable to various open-source LLMs, broadening their potential applications.

6 Limitations

Despite MindRef’s promising potential for recalling reference passages in knowledge-sensitive tasks, it faces limitations in surpassing state-of-the-art retrieval models, especially fine-tuned ones, and struggles with documents lacking clear identifiers or those infrequently encountered during pre-training. Further research is needed to improve instruction tuning for constrained recall and to develop more efficient methods for injecting new knowledge and updating documents without extensive retraining.

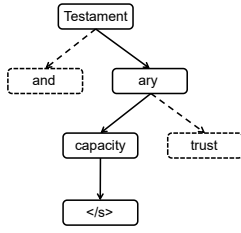
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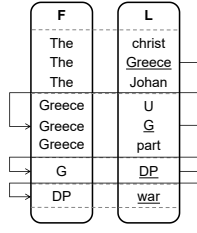
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A Preliminary

A.1 Trie



(a) Title Generation with Prefix Tree.



(b) Passage Prefix Generation with FM-index.

Figure 3: Constrained Decoding Methods: (a) Shows the process of an LLM generating title identifiers using a prefix tree. (b) Shows the process of an LLM generating passage prefixes in a document set via FM-index.

The Trie (Cormen et al., 2022), also known as a dictionary tree or prefix tree, is a tree-like data structure used to store an associative array where the keys are usually strings. Unlike a binary search tree, keys in a Trie are not stored directly within the nodes; instead, they are determined by the node’s position in the tree. All descendants of a node have the same prefix, associated with the string corresponding to that node.

The overall process during constrained decoding using a Trie is shown in Figure 3a. Taking the generation of the title "Testamentary Capacity" as an example, the LLM first selects "Testament" from the set of token strings that start all titles. Subsequently, we can obtain the set of token strings {and, ary} following the string "Testament". After the LLM selects "ary", we get the prefix string "Testamentary", and finally continue to select new strings from the next set of token strings until the end-of-sequence token </s> is encountered, ceasing generation.

A.2 FM-Index

The FM-index (Ferragina and Manzini, 2000) is a data structure used for text retrieval that can store text efficiently with linear space complexity and support fast substring search operations. It is constructed based on the Burrows-Wheeler Transform (BWT) (Burrows et al., 1994). BWT is a method that converts a string into a form that is easy to compress. Given a string, BWT produces a transformed string through the following steps: generate all cyclic shifts of the string, sort all these shifts lexicographically, take the last character of each

sorted shifted string to form a new string, which is the BWT result. For example, for the string "CABAC", the process of building the FM-index is as follows:

F					L
\$ ⁶	C	A	B	A	C ⁵
A ²	B	A	C	\$	C ¹
A ⁴	C	\$	C	A	B ³
B ³	A	C	\$	C	A ²
C ⁵	\$	C	A	B	A ⁴
C ¹	A	B	A	C	\$ ⁶

where \$ is a special string termination token, the numbers in the upper right corner of the letters in the F and L columns are the corresponding position index numbers. The FM-index explicitly stores two main parts: the F column and the L column. The F column is the lexicographically sorted characters of the transformed string, and the L column is the result of BWT. In addition, it stores additional position information to recover the original string from the BWT result. When we want to query a substring, the FM-index starts from the last character of the substring, using the information in the F column and the L column to gradually narrow down the possible position range until the exact position of the substring is determined or the substring is determined to be non-existent.

The overall process during constrained decoding using FM-index is shown in Figure 3b. Considering the generated prefix "The Greece GDP warrants are not technically bonds as investors do" for example, it first starts from the string "The" generated from all corpus, and gets its corresponding L column string set {christ, Greece, Johan}. After "Greece" is selected by the LLM, we can get the next set {U, G, part}, and continue the iteration until reaching the set maximum prefix length to stop generating.

B Additional Details for Experiments

B.1 Datasets

We conduct extensive experiments on 6 knowledge-sensitive tasks from the KILT benchmark (Petroni et al., 2021). These tasks include open-domain QA tasks such as NQ (Kwiatkowski et al., 2019), TriviaQA (Joshi et al., 2017), HotpotQA (Yang et al., 2018), and ELI5 (Fan et al., 2019), the fact-checking task FEVER (Thorne et al., 2018), and the open-domain dialogue system WoW (Dinan et al., 2019). All experiments are tested using the public validation set as divided in the official KILT. Additional details of the datasets are presented in Table

Dataset	Task	Input Format	Output Format	Size
NQ (Kwiatkowski et al., 2019)	Open-domain QA	Question	Extractive	2837
HotpotQA (Yang et al., 2018)	Open-domain QA	Question	Short Abstractive	5600
TriviaQA (Joshi et al., 2017)	Open-domain QA	Question	Extractive	5359
ELI5 (Fan et al., 2019)	Open-domain QA	Question	Long Abstractive	1507
FEVER (Thorne et al., 2018)	Fact Checking	Claim	Classification	10444
WoW (Dinan et al., 2019)	Dialogue	Conversation	Long Abstractive	3054

Table 4: Additional details of the datasets.

4. All the data used in this paper come from the KILT benchmark (Petroni et al., 2021), and KILT is MIT licensed¹. We evaluate the quality of coarse-grained pages and fine-grained reference passage, as well as the enhancement of reference for downstream tasks.

B.2 Prompt

In this subsection, we introduce the prompt used in the first stage for recalling coarse-grained title identifiers, in the second stage for recalling fine-grained reference passage, and in downstream tasks.

B.2.1 Prompt for the First Stage

- Open-domain QA: "Question: {} \n \n The Wikipedia article corresponding to the above question is:\n \n Title:"
- Fact Verification: "Claim: {} \n \n The Wikipedia article corresponding to the above claim is:\n \n Title:"
- Open-domain Dialogue System: "Conversation: {} \n \n The Wikipedia article corresponding to the above conversation is:\n \n Title:"

B.2.2 Prompt for the Second Stage

- Open-domain QA: "Question: {} \n \n The Wikipedia paragraph to answer the above question is:\n \n Answer:"
- Fact Verification: "Claim: {} \n \n The Wikipedia paragraph to support or refute the above claim is:\n \n Answer:"
- Open-domain Dialogue System: "Conversation: {} \n \n The Wikipedia paragraph to answer the above conversation is:\n \n Answer:"

¹<https://opensource.org/licenses/MIT>

B.2.3 Prompt for Reading Comprehension

- Open-domain QA (NQ, TriviaQA, HotpotQA): "Refer to the passage below and answer the following question with just a few words.\n Passage: {} \n Q: {} \n A: The answer is"
- Open-domain QA (ELI5): "Refer to the passage below and answer the following question in detail.\n Passage: {} \n Q: {} \n A:"
- Fact Verification: "background: {} \n claim: {} \n Q: Is the claim true or false? \n A:"
- Open-domain Dialogue System: "background: {} \n {} \n "

B.3 Ablation Study

In this subsection, we conduct ablation studies, comparing methods without weighted scores (w/o weight), without Short Prefix Recall Location (w/o SPRL), and without the first stage of document title recall (w/o first stage). The results are shown in Table 5.

Without weighted scores, relying solely on the recall scores from the second stage leads to a simultaneous decrease in performance for both coarse and fine-grained results, emphasizing the importance of considering scores from both stages. The model, by taking into account title scores, is more capable of selecting the correct document, and within the correct document, it is more likely to choose the correct reference. More results on the choice of weighted α can be found in Figure 4.

Without SPRL, recalling longer segments has a minor impact on page-level performance. However, it significantly affects the quality of fine-grained reference passage, where longer recall lengths paradoxically lead to decreased performance. This result is somewhat counterintuitive and might be due to all document knowledge being stored in the parameters during the pre-training phase, with a short

Method	NQ	TriviaQA	HotpotQA	NQ	TriviaQA	HotpotQA
	R-Precision			Answer in Context		
MindRef	57.77	54.41	48.70	40.82	68.20	30.04
w/o weight	51.22	49.23	48.70	39.06	66.86	28.88
w/o SPRL	55.30	51.50	48.70	37.43	64.64	26.18
w/o first stage	32.22	24.87	23.36	36.27	63.33	24.16

Table 5: Ablation study results on the NQ, TriviaQA, and HotpotQA datasets are shown, with the left half showing the R-Precision at the coarse-grained page level and the right half showing Answer in Context for fine-grained passage. We compared the performance differences without weighted scores, without SPRL, and without the first stage.

Method	NQ	TriviaQA	HotpotQA	NQ	TriviaQA	HotpotQA
	R-Precision			Answer in Context		
MindRef (LLaMA-7b)	54.46	57.03	44.56	36.87	58.48	25.55
MindRef (Vicuna-1.3-7b)	48.47	47.99	40.79	35.28	56.41	23.75
MindRef (LLaMA-13b)	54.42	55.53	46.30	37.72	60.96	26.34
MindRef (Vicuna-1.3-13b)	52.73	46.61	43.41	36.55	67.29	26.63
MindRef (LLaMA-2-7b)	56.33	56.43	46.20	38.07	62.88	27.55
MindRef (LLaMA-2-chat-7b)	3.31	1.12	0.98	4.09	3.97	3.43
MindRef (Vicuna-1.5-7b)	50.76	51.73	41.23	34.16	55.98	24.43
MindRef (LLaMA-2-13b)	57.77	54.41	48.70	40.82	68.20	30.04
MindRef (LLaMA-2-chat-13b)	1.94	1.60	1.55	6.38	7.93	4.71
MindRef (Vicuna-1.5-13b)	52.24	56.34	45.90	37.22	63.24	27.14

Table 6: On the NQ, TriviaQA, and HotpotQA datasets, experimental results after general fine-tuning of the model are presented. The left side shows the page-level R-Precision, while the right side displays the passage-level Answer in Context.

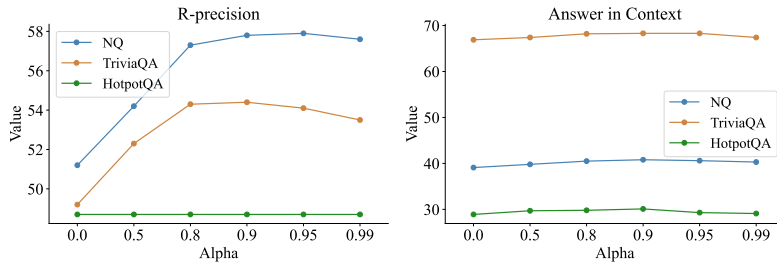


Figure 4: On the NQ, TriviaQA, and HotpotQA datasets, the page-level and passage-level experimental results for LLaMA-2-13b when setting α to $\{0.0, 0.5, 0.8, 0.9, 0.95, 0.99\}$.

prefix sufficient to locate the required reference. Longer references introduce redundancy and noise, thus lowering effectiveness. Notably, when using LLaMA-2-13b for recall, recalling complete passages on the NQ dataset takes about 600 minutes, while recalling short prefixes only requires 150 minutes, significantly reducing time costs. However, considering that dense retrieval takes about 20 minutes, further optimization of speed remains

crucial. More experiments on prefix length can be found in Figure 5.

Without the first stage of document title recall, the quality of reference further declines, significantly impacting the quality of page retrieval. This indicates that using LLMs to directly recall references across a vast number of documents has considerable limitations and opportunities for improvement. The ability of merely prompting LLMs

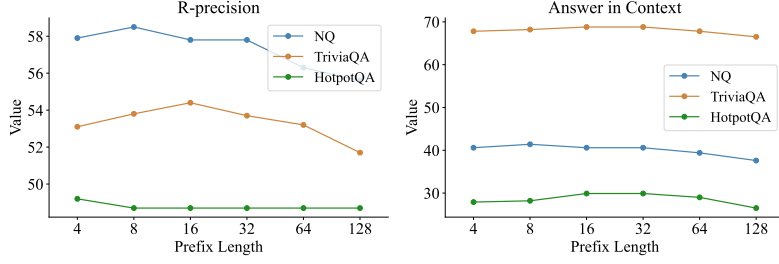


Figure 5: On the NQ, TriviaQA, and HotpotQA datasets, the page-level and passage-level experimental results for LLaMA-2-13b with different prefix token lengths l_{ps} set to $\{4, 8, 16, 32, 64, 128\}$.

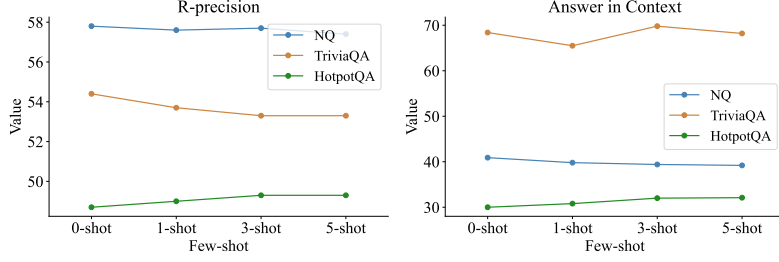


Figure 6: On the NQ, TriviaQA, and HotpotQA datasets, the page-level and passage-level experimental results for LLaMA-2-13b under $\{0, 1, 3, 5\}$ -shot few-shot prompt.

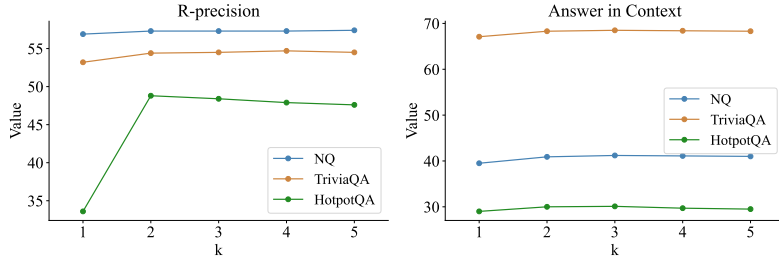


Figure 7: On the NQ, TriviaQA, and HotpotQA datasets, the page-level and passage-level experimental results for LLaMA-2-13b with the number of documents selected in the first stage k set to $\{1, 2, 3, 4, 5\}$.

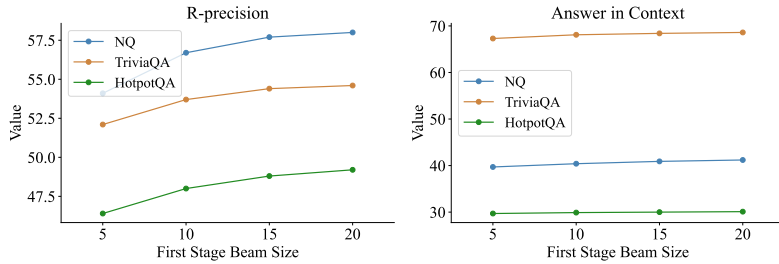


Figure 8: On the NQ, TriviaQA, and HotpotQA datasets, the page-level and passage-level experimental results for LLaMA-2-13b with different beam search sizes $\{4, 8, 16, 32, 64, 128\}$ set for the first stage.

to recall and locate fine-grained reference passage is very limited, making the first stage of recalling document title identifiers crucial.

B.4 Further Analysis

After General Fine-tuning of LLMs. We also test the Vicuna model (Chiang et al., 2023) and the LLaMA-2-chat model refined through rein-

forcement learning from human feedback (Touvron et al., 2023b), both of which underwent general fine-tuning. This general fine-tuning did not significantly enhance the performance of LLMs in recalling and locating reference. This may be due to the paradigm difference between the fine-tuning data and the recall location task, coupled with the fact that most knowledge was already acquired during

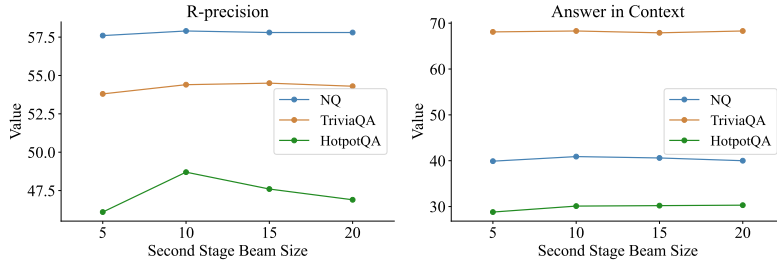


Figure 9: On the NQ, TriviaQA, and HotpotQA datasets, the page-level and passage-level experimental results for LLaMA-2-13b with different beam search sizes {4,8,16,32,64,128} set for the second stage.

the pre-training phase. By creating more diverse recall instruction tuning data, further improvements in model performance might be achieved. Detailed results can be found in Table 6.

Impact of Few-Shot. We explore adding few-shot prompt in the second stage of fine-grained recall and observed its impact on overall performance. This approach brought slight improvements only in the HotpotQA dataset, while showing a slight decline in NQ and TriviaQA. Importantly, adding more few-shot examples significantly reduced generation speed. This suggests that, although few-shot prompting offers a potential path for improvement, extensive exploration is still needed to devise more effective prompting methods. Detailed results can be found in Figure 6.

Impact of Document Selection (k) of First Stage Documents. In Figure 7, we conduct experiments to compare the effect of selecting different numbers of first-stage documents (denoted as k) in MindRef when implemented based on LLaMA-2-13b. We observe that the impact of k on the final performance is not significant, as the necessary effective reference passages are usually contained within the first few documents. The suboptimal performance observed on the HotpotQA dataset when $k = 1$ can be attributed to the dataset requiring two documents to calculate R-Precision.

Impact of Beam Search Sizes. Figures 8 and 9 show the impact of setting different beam sizes in the first and second stages, respectively, in MindRef when implemented based on LLaMA-2-13b. For the first stage of recalling title identifiers, a larger beam size can achieve better page-level results, thereby slightly improving the effectiveness of the second stage of passage recall. However, in the second stage of fine-grained passage recall, the improvement brought by a larger beam size is not significant and may even lead to a slight decline, possibly due to the introduction of additional noise

by a larger beam size.

Memory Usage Analysis. Dense retrieval methods such as Contriever and DPR require over 60GB of memory usage. In contrast, sparse retrieval methods use far less memory, only needing 17GB. The MindRef framework, utilizing FM-index and Trie indexing, requires only 8GB when pre-encoding and storing all documents with FM-index, and the Trie storing all title identifiers needs just 25MB, which is negligible. Compared to sparse and dense retrieval methods, the recall framework effectively saves memory.

B.5 Case Study

In Supplementary Material Tables 7 to 12, we present reference cases obtained using the Gold Standard, BM25, and the MindRef framework with LLaMA-2-13b on the NQ, TriviaQA, and HotpotQA datasets. By generating passages more aligned with the question, MindRef achieves results that contain the answer in Supplementary Material Tables 7, 9, and 11. Supplementary Material Table 8 showcases a biology question; although the passage recalled and located by MindRef does not contain the annotated answer, it provides a more detailed description of the location and process of pancreatic enzyme cleavage of peptide bonds. However, Supplementary Material Tables 10 and 12 show instances where MindRef’s recall failed. This is because merely generating a relevant prefix sometimes cannot ensure that the subsequent part will definitely contain the answer, leading to passages that are only broadly related. Ensuring the flexibility of recall and location while considering more subsequent information still requires further exploration. Nevertheless, we can also note that the references obtained by LLM recall are more natural and easier to read compared to those with predefined segmented beginnings. Finally, compared to the NQ dataset, questions in the TriviaQA and Hot-

763 potQA datasets are longer, allowing for more vo-
764 cabulary overlap between questions and passages,
765 which enhances the effectiveness of BM25 sparse
766 retrieval.

Query: what is the legal age for marriage in australia

Gold Standard: Part II (s.10-21) deals with the marriageable age and the marriage of minors. In the original 1961 Act, marriageable age was set at 16 for females and 18 for males. However, under section 12 of the original 1961 Act a female 14 or 15 years or a male 16 or 17 years could apply to the court for permission to marry. The marriageable age was equalised in 1991 by the "Sex Discrimination Amendment Act 1991", which raised the marriageable age of females to 18.

BM25: marriage. Before 2004, there was no definition of marriage in the 1961 Act, and instead the common law definition used in the English case "Hyde v Hyde" (1866) was considered supreme. Though s.46(1) of the Act required celebrants to explain the legal nature of marriage in Australia to a couple as "the union of a man and a woman to the exclusion of all others, voluntarily entered into for life", these words were descriptive or explanatory, rather than outlining what constituted a legally valid marriage in

MindRef: Marriage in Australia is regulated by the federal "Marriage Act 1961" (Cth), which applies uniformly throughout Australia (including its external territories) to the exclusion of all state laws on the subject. Australian law recognises only monogamous marriages, being marriages of two people, including same-sex marriages, and does not recognise any other forms of union, including traditional Aboriginal marriages, polygamous marriages or concubinage. The marriage age for marriage in Australia is 18 years, but in "unusual and exceptional circumstances" a person aged 16 or 17 can marry with parental consent

Table 7: In the NQ dataset, the Gold Standard, BM25, and MindRef reference passage for the query "what is the legal age for marriage in Australia" are provided. The parts containing the answer are highlighted with a grey background.

Query: where does cleavage of the peptide bond by chymotrypsin occur

Gold Standard: "In vivo", chymotrypsin is a proteolytic enzyme (serine protease) acting in the digestive systems of many organisms. It facilitates the cleavage of peptide bonds by a hydrolysis reaction, which despite being thermodynamically favorable, occurs extremely slowly in the absence of a catalyst. The main substrates of chymotrypsin are peptide bonds in which the amino acid N-terminal to the bond is a tryptophan, tyrosine, phenylalanine, or leucine. Like many proteases, chymotrypsin also hydrolyses amide bonds "in vitro", a virtue that enabled the use of substrate analogs such as N-acetyl-L-phenylalanine p-nitrophenyl amide for enzyme assays.

BM25: 149, producing α -chymotrypsin (which is more active and stable than π -chymotrypsin). The resulting molecule is a three-polypeptide molecule interconnected via disulfide bonds. "In vivo", chymotrypsin is a proteolytic enzyme (serine protease) acting in the digestive systems of many organisms. It facilitates the cleavage of peptide bonds by a hydrolysis reaction, which despite being thermodynamically favorable, occurs extremely slowly in the absence of a catalyst. The main substrates of chymotrypsin are peptide bonds in which the amino acid N-terminal

MindRef: Chymotrypsin cleaves peptide bonds by attacking the unreactive carbonyl group with a powerful nucleophile, the serine 195 residue located in the active site of the enzyme, which briefly becomes covalently bonded to the substrate, forming an enzyme-substrate intermediate. Along with histidine 57 and aspartic acid 102, this serine residue constitutes the catalytic triad of the active site. These findings rely on inhibition assays and the study of the kinetics of cleavage of the aforementioned substrate, exploiting the fact that the en

Table 8: In the NQ dataset, the Gold Standard, BM25, and MindRef reference passage for the query "where does cleavage of the peptide bond by chymotrypsin occur" are provided. The parts containing the answer are highlighted with a grey background.

Query: Which expression is associated with the sinking of the HMS Birkenhead at Gansbaai near Cape Town, South Africa, in February 1852?

Gold Standard: Only 193 of the estimated 643 people on board survived, and the soldiers' chivalry gave rise to the unofficial "women and children first" protocol when abandoning ship, while the "Birkenhead drill" of Rudyard Kipling's poem came to describe courage in face of hopeless circumstances.

BM25: HMS "Birkenhead, also referred to as HM Troopship "Birkenhead or Steam Frigate "Birkenhead", was one of the first iron-hulled ships built for the Royal Navy. She was designed as a steam frigate, but was converted to a troopship before being commissioned. She was wrecked on 26 February 1852, while transporting troops to Algoa Bay at Danger Point near Gansbaai, 87 miles (140 kilometres) from Cape Town in the Cape Colony. There were not enough serviceable lifeboats for all the passengers, and the soldiers

MindRef: The sinking of the "Birkenhead" is one of the earliest maritime disaster evacuations during which the concept of "women and children first" is known to have been applied. "Women and children first" subsequently became standard procedure in relation to the evacuation of sinking ships, in fiction and in life. The term "Birkenhead drill" became defined as courageous behaviour in hopeless circumstances and appeared in Rudyard Kipling's 1893 tribute to the Royal Marines, "Soldier an' Sailor Too": To take your chance in the thick of a rush, with firing all about, Is nothing so bad when you've

Table 9: In the TriviaQA dataset, the Gold Standard, BM25, and MindRef reference passage for the query "Which expression is associated with the sinking of the HMS Birkenhead at Gansbaai near Cape Town, South Africa, in February 1852?" are provided. The parts containing the answer are highlighted with a grey background.

Query: Who was the first winner of 'I'm A Celebrity, Get Me Out Of Here'?

Gold Standard: The first series of I'm a Celebrity...Get Me Out of Here! was broadcast on ITV from 25 August to 8 September 2002. Ant & Dec presented the main show on ITV, whilst Louise Loughman hosted the spin-off show "I'm a Celebrity...Get Me Out of Here! NOW!" on ITV2. The winner of this series was radio DJ Tony Blackburn.

BM25: The first series of I'm a Celebrity...Get Me Out of Here! was broadcast on ITV from 25 August to 8 September 2002. Ant & Dec presented the main show on ITV, whilst Louise Loughman hosted the spin-off show "I'm a Celebrity...Get Me Out of Here! NOW!" on ITV2. The winner of this series was radio DJ Tony Blackburn. The show began with 8 celebrity contestants. The contestants take part in daily trials to earn food All ratings are taken from the UK

MindRef: I'm a Celebrity...Get Me Out of Here! is a British reality TV series in which a number of celebrities live together in a jungle environment for a number of weeks, competing to be crowned "King" or "Queen of the Jungle". The show was originally created in the United Kingdom by the factual programmes department of ITV's then London franchise, London Weekend Television and developed by a team including James Allen, Natalka Znak, Brent Baker and Stewart Morris. The first episode aired on 25 August 2002. It is now produced by ITV Studios and has been licensed globally to countries including the United States

Table 10: In the TriviaQA dataset, the Gold Standard, BM25, and MindRef reference passage for the query "Who was the first winner of 'I'm A Celebrity, Get Me Out Of Here'?" are provided. The parts containing the answer are highlighted with a grey background.

Query: 2014 S/S is the debut album of a South Korean boy group that was formed by who?

Gold Standard: 2014 S/S is the debut album of South Korean group WINNER. It was released on August 12, 2014 by the group's record label, YG Entertainment . The members were credited for writing the lyrics and composing the majority of the album's songs.

BM25: S is a South Korean project group consisting of three members: Kangta, Lee Ji-hoon and Shin Hye-sung. The group debuted in 2003, under the SM Entertainment label. After 11 years, they released and promoted another mini-album in 2014. In 2003, S released their first album "Fr.In.Cl", which stands for "Friends in Classic". In 2014, after more than a decade in hiatus, the group released their second mini-album "Autumn

MindRef: 2014 S/S is the debut album of South Korean group WINNER. It was released on August 12, 2014 by the group's record label, YG Entertainment . The members were credited for writing the lyrics and composing the majority of the album's songs. The members produced the majority of the music for this album themselves, with the help of other producers such as Choice 37, B.I., Airplay, and others. The album was highlighted for incorporating elements generally absent from K-pop releases, including hints of acoustic

Table 11: In the HotpotQA dataset, the Gold Standard, BM25, and MindRef reference passage for the query "2014 S/S is the debut album of a South Korean boy group that was formed by who?" are provided. The parts containing the answer are highlighted with a grey background.

Query: Who is the fictional head of a British Secret Service division and for which a one-time missionary was the inspiration for?

Gold Standard: Charles Fraser-Smith (26 January 1904 – 9 November 1992) was an author and one-time missionary who is widely credited as being the inspiration for Ian Fleming's James Bond quartermaster Q . During World War II, Fraser-Smith worked for the Ministry of Supply, fabricating equipment nicknamed "Q-devices" (after Q-ships) for SOE agents operating in occupied Europe. Prior to the war, Fraser-Smith had worked as a missionary in North Africa. After the war he purchased a dairy farm in Bratton Fleming, Devon, where he died in 1992.

BM25: Q is a fictional character in the James Bond films and film novelisations. Q (standing for Quartermaster), like M, is a job title rather than a name. He is the head of Q Branch (or later Q Division), the fictional research and development division of the British Secret Service. The use of letters as pseudonyms for senior officers in the British Secret Intelligence Service was started by its first director Captain Sir Mansfield George Smith-Cumming (1859-1923) who signed himself with a C written in

MindRef: Ian Fleming created the fictional character of James Bond as the central figure for his works. Bond is an intelligence officer in the Secret Intelligence Service, commonly known as MI6. Bond is known by his code number, 007, and was a Royal Naval Reserve Commander. Fleming based his fictional creation on a number of individuals he came across during his time in the Naval Intelligence Division and 30 Assault Unit during the Second World War, admitting that Bond "was a compound of all the secret agents and commando types I met during the war". Among those types were his brother, Peter, who had been involved in behind-the-lines operations in Norway and Greece during the war.

Table 12: In the HotpotQA dataset, the Gold Standard, BM25, and MindRef reference passage for the query "Who is the fictional head of a British Secret Service division and for which a one-time missionary was the inspiration for?" are provided. The parts containing the answer are highlighted with a grey background.